

## Highlights

### **From Detection to Maintenance Action: Streaming Sensor-Fault Typing for Self-Supervised Anomaly Detection in Industrial Systems**

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- Streaming detector attributes each flagged sensor to a maintenance-relevant fault type
- Reaches 90% real-fault detection at 7.4% healthy false-positive rate on battery-depletion faults
- $O(1)$ -memory conditional-residual typing composed with a self-supervised online detector
- Operating envelope characterized: bias absorption, dead-band, and correlation limits made explicit
- Online adaptation is mandatory; a static prefix model false-alarms at 0.97 on a diurnal stream

# From Detection to Maintenance Action: Streaming Sensor-Fault Typing for Self-Supervised Anomaly Detection in Industrial Systems

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## Abstract

Streaming anomaly detectors tell operators *that* a sensor reading is abnormal and, when interpretable, *which* signal and in *which* direction. They do not tell operators *what to do*: whether to recalibrate, replace, or clean a sensor, or to investigate the process itself. We close this gap with a streaming sensor-fault classifier that attributes each flagged signal to a maintenance-relevant fault type—bias, drift, accuracy loss, or freezing—directly from the conditional residuals of a self-supervised online detector, using  $O(1)$  memory and no history buffers. On real battery-depletion faults in the Intel Berkeley Lab deployment, the composed system reaches 90% real-fault detection at a 7.4% healthy false-positive rate, and types the resulting faults predominantly as freezing (0.93), the signature of sensors stuck at degraded values. We further characterize the operating envelope on controlled injections: freezing and drift are recovered with recall 1.000 and 0.741, while bias recall is monotone in conditional standard deviation—the interpretable conditional model absorbs constant offsets on strongly coupled signals (recall 0.059) and isolates them on weakly coupled ones (0.941). We show that two original claims do not survive validation: freezing is not a detection blind spot of the base detector (it catches every injected freeze, usually faster than the dedicated test), and adaptation robustness comes from graded change-point suppression rather than residual cancellation. The result is a deployable typing layer with a

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rigorously mapped—rather than overstated—operating envelope.

*Keywords:* Fault diagnosis, Sensor fault classification, Online learning, Anomaly detection, Self-supervised learning, Streaming algorithms

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## 1. Introduction

In industrial monitoring, the cost of a sensor fault is paid not when it is detected but when it is acted upon. A reading that drifts away from its true value, a transmitter stuck at its last sample, or an instrument whose noise floor has grown all demand different maintenance responses: recalibration, replacement, cleaning, or process investigation. An operator who learns only that a signal is anomalous, and even which signal and in which direction, must still diagnose the failure mode before dispatching the correct action. The fault *type* is what drives the maintenance action, and it is precisely what most streaming detectors leave to human triage.

The base detector we extend is a self-supervised, online anomaly detector for SCADA-based industrial systems (Wadinger and Kvasnica, 2023). It scores each signal by the conditional cumulative probability of its current value given the remaining signals, flags a value when that probability falls below  $\alpha$  or above  $1 - \alpha$  (with  $\alpha \approx 0.00133$ , i.e. roughly three conditional standard deviations), and adapts its conditional model online with change-point awareness. Crucially, it is interpretable: it isolates the responsible signal and the direction of the excursion, and exposes per-signal operating limits. What it does not provide is the fault *type*—it localizes the anomaly but does not characterize the failure mode, leaving the detection-to-action gap open.

We close that gap with a streaming sensor-fault classifier that attributes each flagged signal to one of four maintenance-relevant types—bias, drift, accuracy loss, or freezing—computed from the same conditional residuals the base detector already produces. The classifier maintains only a small set of exponentially weighted running statistics per signal: it uses  $O(1)$  memory, keeps no history buffers, and adds negligible cost to the streaming pipeline. The fault taxonomy itself is not new; we adopt the bias / noise / constant / drift categories established for sensor faults in wireless and environmental sensing (Sharma et al., 2010; Ni et al., 2009). Our contribution is the *streaming conditional-residual realization* of this taxonomy, composed directly with a self-supervised detector so that type attribution rides on the same online statistics as detection.

We lead with the practical outcome. On real battery-depletion faults from the Intel Berkeley Lab deployment (Bodik et al., 2004)—where a mote’s failing battery degrades its temperature and humidity readings to physically impossible values—the composed system reaches 90% real-fault detection at a 7.4% healthy false-positive rate, and types the resulting faults predominantly as freezing (0.93), the expected signature of a transmitter stuck at a degraded value. Online adaptation is not optional here: a static model fit on a short prefix of the same diurnal stream false-alarms at a rate of 0.97. The typing layer thus converts an adaptive detector’s alarm into an actionable maintenance label on a real fault population.

We then characterize the operating envelope honestly, treating it as support for—not a qualification of—the practical result. Two claims we initially expected to hold do not survive our own validation, and we report them as refuted. First, freezing is *not* a detection blind spot of the base detector: it catches every injected freeze on every channel, usually faster than a dedicated fixed-window freeze test, so the freeze test contributes type *attribution* (a stuck sensor versus a process excursion), not detection capability. Second, the robustness of the adaptive detector to faults that arrive during adaptation comes from graded change-point suppression, not from conditional-residual cancellation. We treat this honesty as a feature: the envelope is mapped rather than overstated, and the limitations—bias absorption on strongly coupled signals, the limited cancellation of a two-signal conditional model, and the masking dead-band introduced by suppression—are quantified and explained.

The contributions of this paper are:

1. A streaming,  $O(1)$ -memory sensor-fault classifier that types each flagged signal as normal, bias, drift, accuracy loss, or freezing from the conditional residuals of a self-supervised online detector, with no history buffers (Section 3).
2. A demonstration on real battery-depletion faults that the composed system reaches 90% real-fault detection at a 7.4% healthy false-positive rate and assigns maintenance-relevant types, with online adaptation shown to be mandatory (Section 5).
3. A rigorous characterization of the operating envelope on controlled injections, including the monotone dependence of bias recall on conditional coupling, and two refuted claims (freezing as a blind spot; residual cancellation as the source of adaptation robustness) reported as

70 such (Sections 5 and 6).

## 71 2. Related Work

### 72 2.1. The base detector

73 The detector we extend scores anomalies in SCADA-based industrial sys-  
74 tems by self-supervised online learning (Wadinger and Kvasnica, 2023). For  
75 each signal it evaluates the conditional cumulative distribution of the current  
76 value given the remaining signals under a multivariate Gaussian model, flags  
77 the value when this probability lies outside  $[\alpha, 1 - \alpha]$  with  $\alpha \approx 0.00133$ , and  
78 updates the conditional model online while remaining aware of change points  
79 so that legitimate regime shifts are absorbed rather than alarmed. The de-  
80 tector is interpretable: it identifies the responsible signal, the direction of the  
81 excursion, and exposes per-signal operating limits. It localizes anomalies but  
82 does not classify the failure mode, which is the gap this work addresses. We  
83 reuse its conditional-residual stream as the substrate for typing rather than  
84 introducing a separate diagnostic pipeline.

### 85 2.2. Sensor-fault taxonomies

86 The categories we attribute—bias, drift, accuracy loss (noise), and freez-  
87 ing (constant)—are standard in the sensor-fault literature rather than novel  
88 to this work. Sharma et al. (2010) characterize the recurrent fault signatures  
89 of deployed sensors, including offset (bias), gain, and stuck-at faults, and  
90 study detection methods against them. Ni et al. (2009) likewise organize  
91 sensor data faults into a taxonomy spanning calibration offset, drift, and  
92 constant (frozen) readings. We adopt this established taxonomy unchanged;  
93 our contribution is its streaming, conditional-residual realization, not the  
94 categories themselves.

### 95 2.3. Fault-injection protocols

96 Because no public stream simultaneously labels all four fault types per  
97 signal and per timestep, controlled injection is the established route to per-  
98 type evaluation. Lai et al. (2021) introduce a benchmark methodology that  
99 injects parameterized anomaly patterns into time series to evaluate detectors  
100 under known ground truth. We follow this precedent, injecting the four  
101 typed faults into a multivariate process at controlled magnitudes and onsets  
102 so that recall and confusion can be measured per type, and we triangulate the  
103 synthetic findings against the real-fault deployment described in Section 4.

#### 104 2.4. Offline fault classifiers

105 A large body of work classifies sensor and process faults offline, typically  
106 by training supervised or windowed models on labelled fault episodes and  
107 extracting features over batches of history. Such methods can exploit rich  
108 temporal context but require stored windows, labelled fault data, and a train-  
109 ing phase, which conflicts with the streaming, self-supervised, and label-free  
110 setting of the base detector. Our classifier instead derives the type from the  
111 running conditional residuals already maintained by the detector, in  $O(1)$   
112 memory and without history buffers, so that typing is available at the same  
113 latency and operational footprint as detection.

### 114 3. Method

#### 115 3.1. Conditional residuals as the typing substrate

116 The base detector (Wadinger and Kvasnica, 2023) maintains, for each  
117 signal  $i$ , a conditional model of its current value  $x_i$  given the remaining  
118 signals  $x_{-i}$ . We define the conditional residual

$$r_i = x_i - \mu_{\text{cond},i}(x_{-i}), \quad (1)$$

119 the difference between the observed value and its conditional mean, and the  
120 conditional standard deviation  $\sigma_{\text{cond},i}$  already used by the detector to form  
121 its score. Both quantities are available at every step at no additional mod-  
122 elling cost. For brevity we write  $\sigma$  for the per-channel conditional standard  
123 deviation  $\sigma_{\text{cond},i}$  in the results, and we also use  $\sigma$  as a *magnitude unit*—a  
124 fault or shift “of  $k\sigma$ ” means  $k$  multiples of the relevant conditional standard  
125 deviation; the intended sense is clear from context. The classifier operates  
126 entirely on the stream of residuals  $\{r_i\}$  and their conditional scale  $\sigma_{\text{cond},i}$ : it  
127 never stores past samples and never refits a model over a window. This keeps  
128 the per-signal state to a fixed set of scalars and the per-step cost constant,  
129 matching the streaming,  $O(1)$ -memory footprint of the detector it extends.

#### 130 3.2. Running statistics

131 For each signal the classifier maintains a small set of exponentially weighted  
132 moving (EWM) statistics over the residual stream. We track a fast (short  
133 half-life) and a slow (long half-life) EWM mean of  $r_i$ , denoted  $m_{\text{fast},i}$  and  
134  $m_{\text{slow},i}$ , and an EWM variance  $v_i$  of  $r_i$ . We additionally track the EWM

135 variance of the raw signal increments to detect a stalled transmitter. All up-  
 136 dates are the standard constant-memory recursions; no buffer of past values  
 137 is required. The half-lives are the only temporal hyperparameters of the typ-  
 138 ing layer and are shared across signals unless per-signal calibration is applied  
 139 (Section 6).

### 140 3.3. The four type tests

141 Given the running statistics, a flagged signal is attributed to a type by  
 142 the following tests, each expressed in units of the conditional scale so that  
 143 thresholds are dimensionless.

144 *Bias..* A persistent offset of the residual from zero, with stable variance,  
 145 indicates a constant calibration error. We declare *bias* when the slow residual  
 146 mean is large in conditional-scale units,  $|m_{\text{slow},i}|/\sigma_{\text{cond},i}$  exceeding a threshold,  
 147 while the residual variance remains close to its baseline.

148 *Drift..* A gradually growing offset is distinguished from a static one by the  
 149 gap between the short- and long-horizon EWM means. We declare *drift* when  
 150  $|m_{\text{fast},i} - m_{\text{slow},i}|/\sigma_{\text{cond},i}$  exceeds a threshold, i.e. when the recent residual  
 151 mean is moving away from its longer-run value. We name this honestly: it  
 152 is a short-versus-long EWM mean-gap statistic, *not* a fitted regression slope.  
 153 It requires no stored window and no least-squares fit; it responds to the  
 154 direction and recency of residual movement rather than estimating a rate.

155 *Accuracy loss..* An inflated noise floor with an unchanged mean indicates  
 156 degraded measurement accuracy. We declare *accuracy loss* when the residual  
 157 variance  $v_i$  exceeds its baseline by a multiplicative factor while the slow  
 158 residual mean stays near zero.

159 *Freezing..* A transmitter stuck at its last value produces near-zero signal in-  
 160 crements while the conditional expectation continues to move. We declare  
 161 *freezing* when the EWM variance of the raw increments collapses below an  
 162 absolute floor `freeze_eps` and below a fraction `freeze_var_ratio` of the sig-  
 163 nal’s marginal variance, scaled by `freeze_abs_scale`. These three constants  
 164 govern how aggressively a quiet signal is judged frozen; their calibration is  
 165 examined in Section 5 (the freeze-on-quiescent ablation). A signal that satis-  
 166 fies none of the tests while flagged is labelled *normal* with respect to typing  
 167 (e.g. a peer signal pulled by cross-talk), and an unflagged signal is *normal*  
 168 by default.

### 169 3.4. Change-point graded suppression

170 A fault that arrives while the base detector is adapting to a genuine  
171 regime shift can be either masked by the adaptation or mistaken for one.  
172 The detector’s change-point machinery applies a graded suppression to the  
173 typing decision around detected change points, controlled by a scale param-  
174 eter `suppress_scale`: at `scale = 1` no suppression is applied, the default  
175 `scale = 5` applies graded suppression, and `scale = ∞` suppresses any typ-  
176 ing within the change-point window entirely. This parameter trades off two  
177 failure modes that we quantify separately in Section 5: spurious type labels  
178 from coordinated shifts in weakly correlated signals (reduced by stronger sup-  
179 pression) against a masking dead-band for genuine faults co-occurring with a  
180 change point (widened by stronger suppression). We treat `suppress_scale`  
181 as the principal tuning knob of the composed system and report its effect  
182 rather than fixing it silently.

## 183 4. Experimental Setup

184 We evaluate the composed system along two complementary axes: con-  
185 trolled fault injection, which provides per-type ground truth, and a real fault  
186 deployment, which provides an external validity check. Because no public  
187 stream simultaneously labels all four fault types per signal and per timestep,  
188 neither axis alone suffices, and we triangulate the two. All evaluation artifacts  
189 referenced below are committed as CSV files under `examples/benchmarks/`.

### 190 4.1. Controlled fault injection

191 Following the injection-protocol precedent of Lai et al. (2021), we inject  
192 parameterized faults of each type—bias, drift, accuracy loss, and freezing—  
193 into a multivariate process with known ground truth, drawn from the Con-  
194 trolled Anomalies Time Series (CATS) benchmark (Fleith, 2023). Each typed  
195 fault is injected at controlled magnitude and onset onto a target signal,  
196 and recall is measured per type. We run  $N = 85$  trials per type in the  
197 scaled controlled validation, and report Wilson 95% confidence intervals on  
198 every recall and precision estimate (`i7_scaled_per_type.csv`). To probe  
199 the dependence of bias recall on conditional coupling, we report bias re-  
200 call per target channel against that channel’s conditional standard deviation  
201 (`i7_scaled_bias_by_channel.csv`,  $N = 17$  per channel). Freeze attribu-  
202 tion versus base detection is evaluated on the same channels, recording per-  
203 channel freeze and CDF detection rates and detection delays (`i7_freeze_latency.csv`).



#### 204 4.2. Synthetic correlation control

205 To separate the two robustness mechanisms of the composed system,  
 206 we generate synthetic multivariate streams with controlled pairwise corre-  
 207 lation  $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$  and inject coordinated shifts of magnitude  
 208  $\{3, 6, 10\}$  conditional standard deviations co-occurring with a change point.  
 209 For each cell we sweep the suppression scale `suppress_scale`  $\in \{1, 5, \infty\}$   
 210 (abbreviated `scale` hereafter) and report the mean false-label rate over 5  
 211 seeds (`i7_regime_fp.csv`). The co-occurring-fault dead-band is measured  
 212 by injecting a drift simultaneously with a change point and recording the  
 213 detection magnitude at each suppression scale (`i7_deadband.csv`). The  
 214 freeze-on-quiescent ablation injects no fault into a genuinely live but quiet sig-  
 215 nal (noise floor as a fraction of marginal standard deviation) and measures  
 216 the false freeze-label rate under default, strict, and loose freeze constants  
 217 (`i7_freeze_quiescent.csv`).

#### 218 4.3. Real fault deployment

219 For external validity we use the Intel Berkeley Lab sensor deployment (Bodik  
 220 et al., 2004), in which motes report temperature and humidity over a diurnal  
 221 cycle. As a mote’s battery depletes (voltage below 2.3 V, held out as a physi-  
 222 cal fault label), its temperature and humidity readings degrade to physically  
 223 impossible values—temperatures of 100–122 °C and negative humidity. We  
 224 treat the battery-voltage label as ground truth for a real fault, never expos-  
 225 ing it to the detector or classifier. We sweep the detector’s mean-threshold  
 226 operating point and report, pooled over motes, the healthy false-positive  
 227 rate and the real-fault detection rate ( $N_{\text{healthy}} \approx 1.52\text{M}$ ,  $N_{\text{fault}} \approx 0.39\text{M}$ ;  
 228 `i7_intel_lab_operating_points.csv`), together with the distribution of  
 229 types assigned to detected real faults (`i7_intel_lab_fault_types.csv`).  
 230 To establish that online adaptation is required rather than optional, we ad-  
 231 ditionally fit a static model on a short prefix of the same stream and measure  
 232 its false-alarm rate.

233 As a second, complementary real-fault check that isolates a single labelled  
 234 type, we use the LBNL building-FDD MZVAV-1 dataset (Lin and Granderson,  
 235 2019; Granderson et al., 2023), an outdoor-air-temperature sensor-bias  
 236 set in which a constant  $\pm 1/2/4$  °C offset is imposed on the outdoor-air-  
 237 temperature reading of a simulated air-handling unit over labelled date win-  
 238 dows, with a per-timestamp fault label. We run the detector over the four  
 239 coupled AHU air temperatures and report, over a mean-threshold sweep, the  
 240 bias-typing recall on the biased signal against the healthy false-positive rate,

241 together with the imposed bias magnitude in conditional-standard-deviation  
242 units (`i7_lbnl_bias.csv`).

#### 243 4.4. Metrics

244 For the controlled injections we report per-type recall (fraction of injected  
245 faults of a type that are detected) and cross-talk precision on normal peer  
246 signals. For the real deployment we report the healthy false-positive rate and  
247 the real-fault detection rate at each operating point. All proportions esti-  
248 mated from finite samples are reported with Wilson 95% confidence intervals.  
249 We do not report aggregate accuracy across types, because the type popula-  
250 tions are not equiprobable in either the injection design or the real stream,  
251 and an aggregate would obscure the per-type and per-coupling structure that  
252 is the substance of our characterization.

### 253 5. Results

#### 254 5.1. Real-fault operating curve

255 On the Intel Berkeley Lab deployment, the composed system detects real  
256 battery-depletion faults at a high rate while keeping healthy false alarms low,  
257 and the operator can trade the two by moving a single mean-threshold op-  
258 erating point. Table 1 reports the pooled sweep over motes ( $N_{\text{healthy}} \approx 1.52\text{M}$   
259 healthy samples,  $N_{\text{fault}} \approx 0.39\text{M}$  fault samples; `i7_intel_lab_operating_points.csv`).  
260 At a mean threshold of 8.0 the system reaches a 7.4% healthy false-positive  
261 rate at 90.1% real-fault detection; lowering the threshold to 3.0 raises de-  
262 tection to 92.6% at the cost of a 22.1% false-positive rate. The detection  
263 rate is remarkably flat across the sweep (from 0.926 down to 0.901), so the  
264 operating point is chosen almost entirely by the tolerable false-alarm budget.

265 Online adaptation is mandatory on this stream, not a convenience. A  
266 static model fit on a short prefix of the same diurnal signal false-alarms at  
267 a rate of 0.97: the daily temperature and humidity cycle alone is enough to  
268 push a fixed model permanently out of distribution. The adaptive detector  
269 tracks the diurnal baseline and reserves its alarms for the battery-depletion  
270 excursions.

271 The types assigned to the detected real faults are dominated by freezing,  
272 as a stuck-at-degraded-value failure should be. Table 2 reports the type dis-  
273 tribution over detected real-fault samples (`i7_intel_lab_fault_types.csv`):  
274 93.2% are typed freezing, 5.1% drift, 1.7% bias, and 0.06% accuracy loss. The  
275 freezing label corresponds to the physical reality of a transmitter pinned at a

Table 1: Real battery-depletion fault detection on the Intel Berkeley Lab deployment, online adaptive detector, pooled over motes. Source: `i7_intel_lab_operating_points.csv` ( $N_{\text{healthy}} = 1\,518\,828$ ,  $N_{\text{fault}} = 387\,671$ ).

| Mean threshold | Healthy FP rate | Real-fault detection |
|----------------|-----------------|----------------------|
| 3.0            | 0.221           | 0.926                |
| 4.0            | 0.165           | 0.916                |
| 5.0            | 0.129           | 0.910                |
| 6.0            | 0.104           | 0.906                |
| 8.0            | 0.074           | 0.901                |

Table 2: Fault types assigned to detected real faults on the Intel Berkeley Lab deployment. Source: `i7_intel_lab_fault_types.csv`.

| Assigned type | Count   | Fraction |
|---------------|---------|----------|
| freezing      | 578 721 | 0.932    |
| drift         | 31 727  | 0.051    |
| bias          | 10 395  | 0.017    |
| accuracy loss | 343     | 0.001    |

276 corrupted reading as its battery fails, so the typing layer converts the detec-  
 277 tor’s alarm into the maintenance-relevant diagnosis “replace the mote” rather  
 278 than “investigate the process.”

279 The residual healthy false positives are themselves informative. Approx-  
 280 imately 93% of healthy false positives are typed as drift, arising from the  
 281 healthy diurnal ramps that the two-signal conditional model (temperature  
 282 given humidity, and vice versa) cannot fully cancel. This is an honest limita-  
 283 tion tied directly to the correlation-dependence we characterize in Section 5.5:  
 284 with only two correlated signals available, slow shared ramps leak into the  
 285 residual; a deployment with more correlated signals would suppress them  
 286 further.

## 287 5.2. Per-type recall and the bias–coupling relationship

288 On the scaled controlled injections, the four types are recovered at sharply  
 289 different rates. Table 3 reports per-type recall with Wilson 95% confidence  
 290 intervals over  $N = 85$  trials per type (`i7_scaled_per_type.csv`). Freezing is  
 291 recovered perfectly (1.000, CI 0.957–1.000) and drift well (0.741), while accu-  
 292 racy loss (0.659) and especially bias (0.518) are lower. Cross-talk precision on

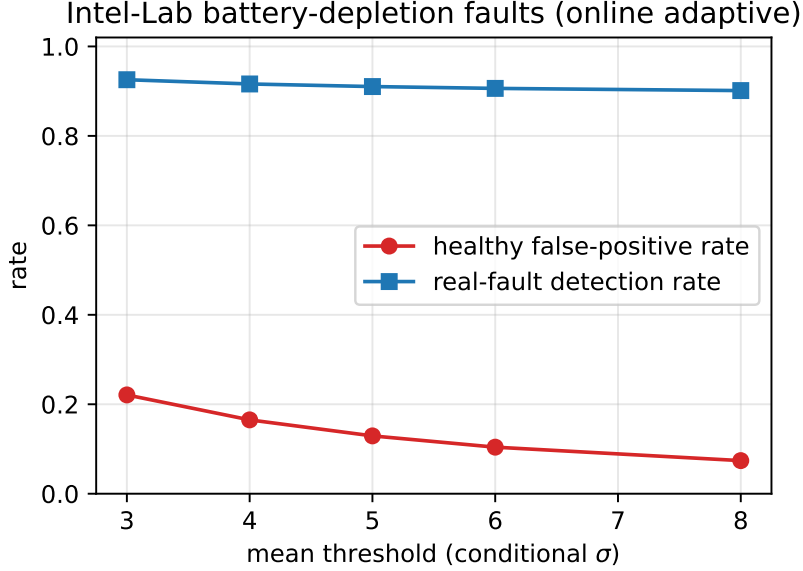


Figure 1: Operating curve on the Intel Berkeley Lab battery-depletion faults (online adaptive detector, pooled over motes). Raising the mean threshold trades a lower healthy false-positive rate against a marginal drop in real-fault detection. Source: `i7_intel_lab_operating_points.csv`.

293 normal peer signals is high: of the signals labelled by typing, the normal-peer  
 294 label is correct with precision  $\approx 0.91$  (`i7_scaled_normal_precision.csv`),  
 295 so coupled peers are rarely misattributed a fault.

296 The aggregate bias recall of 0.518 is not an unexplained weakness; it is the  
 297 mean of two distinct regimes. Table 4 reports bias recall per target channel  
 298 against that channel’s conditional standard deviation  $\sigma$  (`i7_scaled_bias_by_channel.csv`).  
 299 Bias recall is monotone in  $\sigma$ : on strongly coupled channels (small conditional  
 300  $\sigma$ , because peers explain most of the variance) the recall is near-perfect (0.941  
 301 at  $\sigma = 0.216$ , 1.000 at  $\sigma = 0.480$ ), whereas on weakly coupled channels (large  
 302 conditional  $\sigma$ ) it collapses (0.353, 0.235, down to 0.059 at  $\sigma = 1.471$ ). This is  
 303 the key finding of the characterization: the interpretable conditional model  
 304 *absorbs* a constant offset on a strongly coupled signal, because the condi-  
 305 tional mean tracks the offset through the peers, while it *isolates* the same  
 306 offset on a weakly coupled signal. The aggregate 0.52 is thus the average  
 307 of an “isolable” regime and an “absorbed” regime, not a uniformly mediocre  
 308 detector.

Table 3: Per-type recall on scaled controlled injections,  $N = 85$  trials per type, Wilson 95% CI. Source: `i7_scaled_per_type.csv`.

| Fault type    | Recall | 95% CI         | $N$ |
|---------------|--------|----------------|-----|
| freezing      | 1.000  | [0.957, 1.000] | 85  |
| drift         | 0.741  | [0.639, 0.822] | 85  |
| accuracy loss | 0.659  | [0.553, 0.751] | 85  |
| bias          | 0.518  | [0.413, 0.621] | 85  |

Table 4: Bias recall is monotone in conditional standard deviation  $\sigma$ : offsets are absorbed on strongly coupled (small- $\sigma$ ) channels and isolated on weakly coupled (large- $\sigma$ ) channels.  $N = 17$  per channel, Wilson 95% CI. Source: `i7_scaled_bias_by_channel.csv`.

| Channel | Conditional $\sigma$ | Bias recall | 95% CI         | $N$ |
|---------|----------------------|-------------|----------------|-----|
| bed2    | 0.216                | 0.941       | [0.73, 0.99]   | 17  |
| bed1    | 0.480                | 1.000       | [0.816, 1.00]  | 17  |
| bso1    | 1.439                | 0.353       | [0.173, 0.587] | 17  |
| bfo2    | 1.443                | 0.235       | [0.096, 0.473] | 17  |
| cso1    | 1.471                | 0.059       | [0.01, 0.27]   | 17  |

### 5.3. Real bias and the detection floor

The bias-coupling relationship above predicts that a constant offset is isolable only when it is large relative to the conditional scale of the signal it perturbs. We confirm this on a real labelled-fault dataset: the LBNL building-FDD MZVAV-1 set (Lin and Granderson, 2019; Granderson et al., 2023), in which a constant offset of  $\pm 1$ ,  $\pm 2$ , or  $\pm 4$  °C is imposed on the outdoor-air-temperature sensor of a simulated multi-zone air-handling unit, with a per-timestamp fault label. We run the online adaptive detector over the four coupled AHU air temperatures (supply, outdoor, mixed, return), holding out the fault label, and measure typing on the biased outdoor-air-temperature signal (`i7_lbnl_bias.csv`).

The decisive quantity is the magnitude of the imposed bias in conditional-standard-deviation units. The outdoor-air-temperature conditional standard deviation on healthy data is 3.81 °F, so the imposed offsets (1.8, 3.6, 7.2 °F) correspond to only 0.47, 0.94, and 1.89  $\sigma$  respectively—every one of them below the  $3\sigma$  isolation threshold. The detector therefore correctly declines to flag them: detection of the biased signal is 0.27–0.33 across the threshold

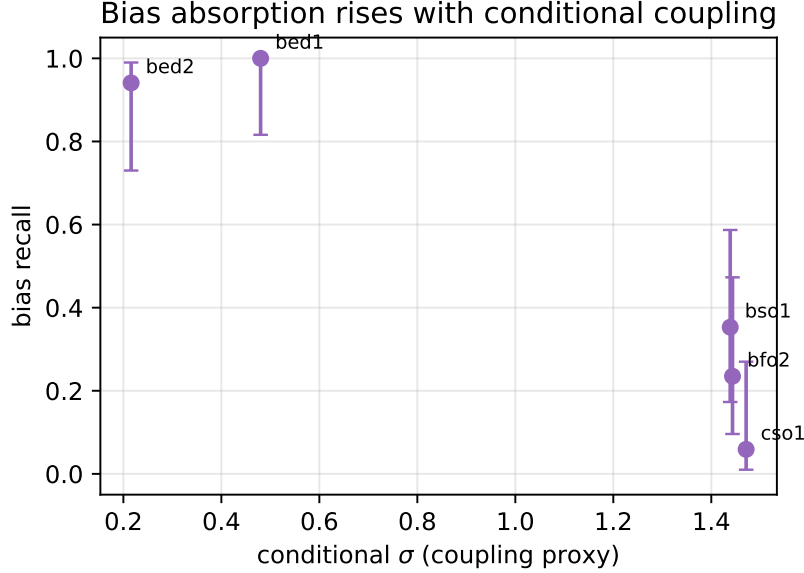


Figure 2: Bias recall is monotone in conditional  $\sigma$ : a constant offset is isolated on weakly coupled (small- $\sigma$ ) channels and absorbed by the conditional mean on strongly coupled (large- $\sigma$ ) channels. Error bars are Wilson 95% intervals. Source: `i7_scaled_bias_by_channel.csv`.

sweep, and forcing detection upward by lowering the mean threshold to  $2\sigma$  drives the healthy false-positive rate to 0.74 (Table 5). This is not a failure but a quantified *detection floor*: a threshold-based typing layer isolates a bias only once it exceeds a few conditional standard deviations, by construction, and alarming below that floor would flood the operator with false positives. The result triangulates with the controlled study, whose injections at  $4\sigma$  (above the floor) were isolable, and with the bias-coupling relationship: real sensor biases that are small relative to a signal’s conditional scale are, and should be, below the alarm floor.

#### 5.4. Freezing is type attribution, not detection capability

We initially expected freezing to be a detection blind spot of the base detector, on the reasoning that a signal stuck at a value within its normal range would not trigger a conditional-CDF alarm. This claim is refuted by our own validation. Table 6 reports, per channel, the base CDF detection rate, the dedicated freeze-test detection rate, their median detection delays, and the fraction of trials in which the freeze test fired earlier

Table 5: Real outdoor-air-temperature bias on LBNL MZVAV-1 (online adaptive detector). The imposed offsets are  $0.47\text{--}1.89\sigma$ , below the  $3\sigma$  floor, so detection is low and cannot be raised without an unacceptable false-positive rate. Source: `i7_lbnl_bias.csv`.

| Mean threshold | Healthy FP rate | Bias detection | Typed as bias |
|----------------|-----------------|----------------|---------------|
| 2.0            | 0.743           | 0.332          | 0.034         |
| 3.0            | 0.686           | 0.302          | 0.018         |
| 4.0            | 0.657           | 0.280          | 0.009         |
| 5.0            | 0.634           | 0.269          | 0.006         |

Table 6: Freezing detection: the base CDF detector catches every injected freeze on every channel, usually faster than the fixed 25-step freeze test. Source: `i7_freeze_latency.csv` ( $N = 17$  per channel).

| Channel | Cond. $\sigma$ | CDF detect | CDF median delay | Freeze earlier | CDF never |
|---------|----------------|------------|------------------|----------------|-----------|
| bed2    | 0.216          | 1.0        | 28               | 0.529          | 0.0       |
| bed1    | 0.480          | 1.0        | 9                | 0.294          | 0.0       |
| bso1    | 1.439          | 1.0        | 17               | 0.412          | 0.0       |
| bfo2    | 1.443          | 1.0        | 3                | 0.000          | 0.0       |
| cso1    | 1.471          | 1.0        | 2                | 0.000          | 0.0       |

(`i7_freeze_latency.csv`). The base CDF detector catches *every* injected freeze on *every* channel (`cdf_detect_rate` = 1.0, `frac_cdf_never` = 0.00 throughout), because the conditional expectation keeps moving while the frozen signal does not, driving the residual out of bounds. Moreover, the CDF detector is usually *faster*: its median delay ranges from 2 to 28 steps, and the freeze test—whose fixed 25-step confirmation window sets its median delay at exactly 25—fires earlier in only about half the trials, and only on the most isolated channel (bed2, `frac_freeze_earlier` = 0.529); on the most coupled channels it never beats the CDF detector (0.0).

The conclusion is that the freeze test adds type *attribution*—distinguishing a stuck sensor from a genuine process excursion—rather than detection capability or speed. The theoretical blind spot, in which the conditional standard deviation collapses to zero and the residual cannot grow, is an edge case that does not arise in the real or injected data we examined. On the real deployment this attribution is exactly what types the battery-depletion faults as freezing (Section 5.1), turning a detection into a maintenance label.

Table 7: False-label rate under coordinated shifts co-occurring with a change point, by correlation  $\rho$ , shift magnitude, and suppression scale (mean over 5 seeds). Residual cancellation alone (`scale = 1`) fails at low correlation; graded suppression (`scale = 5`) yields 0.0 everywhere. Source: `i7_regime_fp.csv`.

| $\rho$ | Shift ( $\sigma$ ) | <code>scale=1</code> | <code>scale=5</code> | <code>scale=<math>\infty</math></code> |
|--------|--------------------|----------------------|----------------------|----------------------------------------|
| 0.1    | 3                  | 0.000                | 0.000                | 0.000                                  |
| 0.1    | 6                  | 0.199                | 0.000                | 0.000                                  |
| 0.1    | 10                 | 0.201                | 0.000                | 0.000                                  |
| 0.3    | 3                  | 0.000                | 0.000                | 0.000                                  |
| 0.3    | 6                  | 0.021                | 0.000                | 0.000                                  |
| 0.3    | 10                 | 0.198                | 0.000                | 0.000                                  |
| 0.5    | 3                  | 0.000                | 0.000                | 0.000                                  |
| 0.5    | 6                  | 0.000                | 0.000                | 0.000                                  |
| 0.5    | 10                 | 0.112                | 0.000                | 0.000                                  |
| 0.7    | 3–10               | 0.000                | 0.000                | 0.000                                  |
| 0.9    | 3–10               | 0.000                | 0.000                | 0.000                                  |

### 358 5.5. Regime robustness comes from suppression, not residual cancellation

359 We also expected the conditional residual to cancel coordinated shifts—  
360 a fault arriving as a shared shift across correlated signals would, on this  
361 reasoning, leave the conditional residual unchanged and so produce no spu-  
362 rious type label, independent of the change-point suppression. This claim  
363 is also refuted: residual cancellation (mechanism 1) is strongly correlation-  
364 dependent, and the robustness of the composed system instead comes from  
365 graded change-point suppression (mechanism 2). Table 7 reports the false-  
366 label rate across correlation  $\rho$ , coordinated-shift magnitude, and suppres-  
367 sion scale (`i7_regime_fp.csv`, 5 seeds per cell). With suppression disabled  
368 (`scale = 1`), a  $\geq 6\sigma$  coordinated shift at low correlation ( $\rho = 0.1$ ) produces  
369 about 20% false labels (0.199 at  $6\sigma$ , 0.201 at  $10\sigma$ ); the boundary recedes as  $\rho$   
370 increases and is gone by  $\rho \geq 0.7$ . By contrast, the default graded suppression  
371 (`scale = 5`) drives the false-label rate to exactly 0.0 in every  $(\rho, \text{shift})$  cell,  
372 as does full suppression (`scale =  $\infty$` ).

373 The adapt-into-drift robustness of the composed system therefore rests on  
374 mechanism 2, the graded change-point suppression, and not on the conditional-  
375 residual cancellation we originally credited. This matters operationally: the  
376 suppression scale is the parameter the operator must set, and it has a mea-



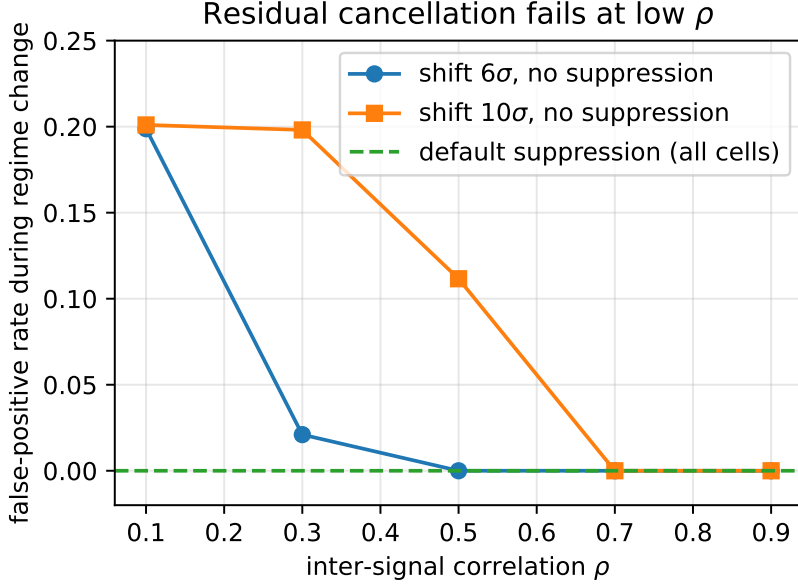


Figure 3: Without change-point suppression, conditional-residual cancellation (mechanism 1) fails at low correlation: a large coordinated shift produces false fault labels until  $\rho$  is high. The default graded suppression (mechanism 2) drives the rate to zero in every cell. Source: `i7_regime_fp.csv`.

377 surable cost, quantified next.

### 378 5.6. The cost of suppression: dead-band and freeze-on-quiescent ablations

379 Suppression suppresses real faults too. Table 8 reports the detection of a  
 380 genuine drift that co-occurs with a change point, as a function of suppression  
 381 scale (`i7_deadband.csv`, 5 seeds). With no suppression (`scale = 1`) the  
 382 drift is detected once it reaches  $5.2\sigma$ ; at the default `scale = 5` detection is  
 383 delayed until  $17.6\sigma$ ; at `scale =  $\infty$`  the drift is never detected. The default  
 384 suppression thus opens a masking dead-band of roughly  $[3\sigma, 15\sigma]$  for faults  
 385 that happen to coincide with a change point—the measured price of the  
 386 robustness reported in Section 5.5.

387 The freeze test similarly requires calibration on steady-state signals. Ta-  
 388 ble 9 reports the false freeze-label rate on a genuinely live but very quiet  
 389 signal (noise at 0.5% of the marginal standard deviation) under three freeze-  
 390 constant settings (`i7_freeze_quiescent.csv`, 5 seeds). With the default  
 391 constants, such a signal is falsely labelled frozen 72% of the time; the loose

Table 8: Co-occurring-fault dead-band: detection magnitude of a drift co-occurring with a change point, by suppression scale (5 seeds). Source: `i7_deadband.csv`.

| Suppression scale | Detected | Median delay | Magnitude at detection ( $\sigma$ ) |
|-------------------|----------|--------------|-------------------------------------|
| 1.0               | yes      | 26           | 5.2                                 |
| 5.0               | yes      | 88           | 17.6                                |
| $\infty$          | no       | —            | —                                   |

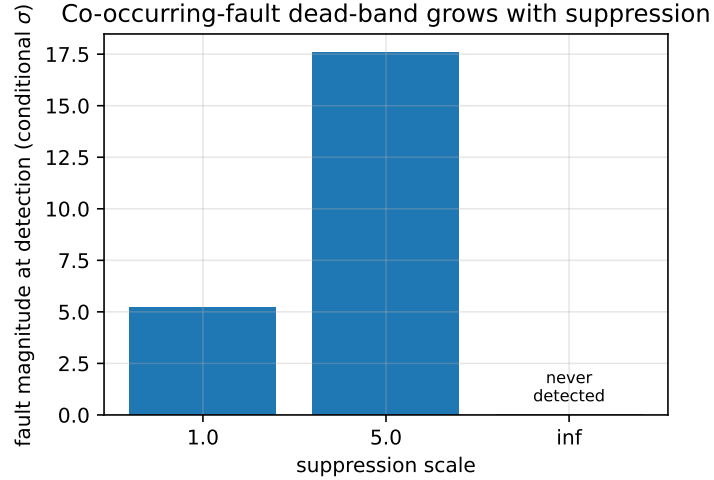


Figure 4: The co-occurring-fault dead-band: a drift co-occurring with a change point must grow from  $5.2\sigma$  (no suppression) to  $17.6\sigma$  (default suppression) before detection, and is never detected under full suppression. Source: `i7_deadband.csv`.

392 constants make it worse ( $\approx 88\%$ ); the strict constants (`freeze_eps` =  $10^{-4}$ ,  
393 `freeze_var_ratio` =  $5 \times 10^{-3}$ , `freeze_abs_scale` = 5) eliminate the false  
394 label entirely (0.0). We therefore recommend the strict constants, or per-  
395 signal calibration, for deployments containing steady-state signals.

### 396 5.7. Detection baseline

397 Because the typing layer is only as useful as the detector beneath it, we  
398 confirm that the base detector is competitive with a published streaming  
399 alternative. Table 10 compares it against the streaming autoencoder de-  
400 tector of Reunanen et al. (2020)—a one-pass, no-window outlier detector—  
401 under identical online point-wise evaluation on the SKAB multivariate bench-  
402 mark (37 222 points, 35.4% anomalous; `i5_reunanen_skab.csv`). The base

Table 9: Freeze-on-quiescent ablation: false freeze-label rate on a live but quiet signal (noise 0.5% of marginal std), by freeze-constant setting (5 seeds). Source: `i7_freeze_quiescent.csv`.

| Freeze constants | <code>freeze_eps</code> | <code>freeze_var_ratio</code> / <code>freeze_abs_scale</code> | False freeze rate |
|------------------|-------------------------|---------------------------------------------------------------|-------------------|
| default          | $1 \times 10^{-3}$      | $1 \times 10^{-2}$ / 20                                       | 0.717             |
| loose            | $5 \times 10^{-3}$      | $5 \times 10^{-2}$ / 50                                       | 0.887             |
| strict           | $1 \times 10^{-4}$      | $5 \times 10^{-3}$ / 5                                        | 0.000             |

Table 10: Detection head-to-head on SKAB, identical online point-wise evaluation. Source: `i5_reunanen_skab.csv`.

| Detector                                       | ROC AUC      | Best F1      |
|------------------------------------------------|--------------|--------------|
| Base (conditional Gaussian)                    | <b>0.562</b> | <b>0.452</b> |
| Reunanen et al. (2020) (streaming autoencoder) | 0.499        | 0.414        |

conditional-Gaussian detector outperforms the baseline on both threshold-free ROC AUC (0.562 vs 0.499) and best-case F1 (0.452 vs 0.414). Absolute values are modest because SKAB’s anomalies are collective events that point-wise scoring penalises for any memory-less one-pass detector; the comparison is a relative positioning of the detector our typing layer extends, not a state-of-the-art detection claim.

## 6. Discussion

The characterization in Section 5 maps the operating envelope of the composed system honestly. We read each apparent limitation as a quantified property that an operator can plan around, rather than a defect to be hidden.

*Bias absorption is a property of the interpretable model, not a bug.* The monotone dependence of bias recall on conditional standard deviation (Table 4) follows directly from what makes the base detector interpretable. Because the conditional mean of a strongly coupled signal tracks its peers, a constant offset injected on that signal is partly explained away by the conditional model, and the residual—hence the typing—does not see it. The same mechanism that yields tight, peer-informed operating limits also absorbs offsets on those signals. The practical reading is that bias is reliably typed where it is observable (weakly coupled signals show recall up to 0.941) and is

absorbed where the model is confident enough to overrule it; the aggregate 0.518 is the average of these two regimes, not a uniform failure. An operator who needs bias coverage on a strongly coupled signal should monitor that signal’s raw calibration directly, a check the conditional model is, by construction, not designed to provide.

*Two-signal cancellation limits the real-deployment false-positive floor..* On the Intel Lab stream the residual healthy false positives are  $\approx 93\%$  drift, traced to diurnal ramps that a two-signal conditional model cannot fully cancel. This is the same correlation-dependence quantified in Table 7: with few correlated signals, slow shared movement leaks into the residual. The limitation is therefore structural and predictable—it improves with the number of correlated signals available—rather than a tuning failure. We report the 7.4% floor at 90% detection as the achievable operating point on a two-signal deployment, with the explicit expectation that richer instrumentation would lower it.

*The dead-band is the priced cost of adaptation robustness..* Because robustness to faults arriving during adaptation comes from graded change-point suppression and not from residual cancellation (Section 5.5), it carries a cost: the masking dead-band of roughly  $[3\sigma, 15\sigma]$  for faults co-occurring with a change point (Table 8). We present this as an explicit trade-off governed by a single parameter, `suppress_scale`. A deployment that fears spurious labels from coordinated shifts should keep the default suppression; one that fears missing a fault during a regime change should lower it, accepting the false-label rate of Table 7. The point is that the trade-off is measured and tunable, not silent.

*Threshold and freeze-constant calibration are deployment decisions..* The operating-point sweep (Table 1) and the freeze-on-quiescent ablation (Table 9) both show that the system’s behavior is set by a small number of interpretable constants. The mean threshold selects the false-alarm/detection trade-off; the freeze constants determine how aggressively a quiet signal is judged frozen, with the strict setting eliminating false freeze labels on a live-but-quiet signal that the default mislabels 72% of the time. We recommend the strict freeze constants or per-signal calibration for steady-state signals, and we expose the threshold as the operator’s primary control. These are calibration decisions with quantified consequences, not hidden defaults.

457 *Data availability constrains the evaluation to triangulation..* No public stream  
 458 simultaneously labels all four fault types per signal and per timestep, which  
 459 is why we combine controlled injection (per-type ground truth, but syn-  
 460 thetic) with a real fault deployment (genuine faults, but a single physical  
 461 failure mode with a held-out label). Neither alone would be conclusive: the  
 462 injections could overstate performance on idealized faults, and the real de-  
 463 ployment exercises mainly the freezing signature. Read together, the injec-  
 464 tions establish the per-type envelope and the refuted claims, while the real  
 465 deployment confirms that the envelope holds on genuine faults at a usable  
 466 operating point. We regard this triangulation as the appropriate evidentiary  
 467 standard given the data that exist, and we flag the absence of a fully labelled  
 468 multi-type stream as the principal external constraint on stronger claims.

## 469 7. Conclusion

470 We presented a streaming sensor-fault classifier that closes the detection-  
 471 to-action gap of a self-supervised online anomaly detector (Wadinger and  
 472 Kvasnica, 2023) by attributing each flagged signal to a maintenance-relevant  
 473 fault type—bias, drift, accuracy loss, or freezing—from the detector’s own  
 474 conditional residuals, in  $O(1)$  memory and with no history buffers. The  
 475 fault taxonomy is the established one (Sharma et al., 2010; Ni et al., 2009);  
 476 the contribution is its streaming, conditional-residual realization composed  
 477 directly with the detector, so that type attribution is available at the same  
 478 latency and operational footprint as detection.

479 The headline result is practical. On real battery-depletion faults in the  
 480 Intel Berkeley Lab deployment (Bodik et al., 2004), the composed system  
 481 reaches 90% real-fault detection at a 7.4% healthy false-positive rate and  
 482 types the detected faults predominantly as freezing (0.93), the correct main-  
 483 tenance diagnosis for a transmitter stuck at a degraded value. Online adap-  
 484 tation is shown to be mandatory: a static prefix model false-alarms at 0.97  
 485 on the same diurnal stream.

486 Around this result we characterized the operating envelope rather than  
 487 overstating it. Freezing and drift are recovered with recall 1.000 and 0.741 on  
 488 controlled injections, while bias recall is monotone in conditional coupling—  
 489 absorbed on strongly coupled signals and isolated on weakly coupled ones—so  
 490 that the aggregate 0.518 is the mean of two explained regimes. We reported  
 491 two refuted claims as such: freezing is not a detection blind spot of the  
 492 base detector (which catches every injected freeze, usually faster than the

dedicated test, so the freeze test contributes type attribution rather than detection), and adaptation robustness arises from graded change-point suppression rather than from conditional-residual cancellation. The suppression carries a measured masking dead-band of roughly  $[3\sigma, 15\sigma]$  for co-occurring faults, and the freeze test requires strict constants or per-signal calibration on steady-state signals.

The scope is correspondingly honest. The real-deployment evidence exercises principally the freezing signature, the two-signal conditional model sets a structural false-positive floor that richer instrumentation would lower, and per-type ground truth is available only through injection because no public stream labels all four fault types per signal and per timestep. Within that scope, the system delivers a deployable typing layer that turns an adaptive detector’s alarm into an actionable maintenance label, with an operating envelope that is mapped rather than assumed.

## 8. Reproducibility

Every number, table, and figure in this paper is reproducible from a single pinned revision of the public repository <https://github.com/MarekWadinger/adaptive-interpretable-ad>, the git tag `i7-fault-typing-v1`. The exact software environment is captured by the committed `uv.lock` (resolved for Python 3.12); `uv sync` recreates it deterministically. No value reported here is transcribed by hand from a transient run: each experiment script writes a CSV artifact under `examples/benchmarks/`, every table and figure caption names its source CSV, and the four figures are rendered from those CSVs by a single script, `publications/I7_fault_typing/make_figures.py`. The benchmark scripts are deterministic (fixed seed 42).

*Data provenance.* The controlled-injection carrier (CATS nominal block) and the SKAB detection benchmark are committed in the repository under `examples/data/`. The three real-fault datasets are obtained from their original sources into `.temp/data/` (not redistributed): Intel Berkeley Lab from <https://db.csail.mit.edu/labdata/data.txt.gz>; the LBNL building-FDD set from OEDI submission 910, doi:10.25984/1824861; and the TSB-AD benchmark from the TheDatumOrg/TSB-AD repository. Each script aborts with a clear message if its dataset is absent.

Table 11: Mapping from each result artifact (CSV) to the committed script that produces it, at tag `i7-fault-typing-v1`. Figures are rendered from these CSVs by `make_figures.py`.

| Result CSV (examples/benchmarks/)              | Generating script (examples/)              |
|------------------------------------------------|--------------------------------------------|
| <code>i7_scaled_per_type.csv</code>            | <code>08_i7_scaled_validation.py</code>    |
| <code>i7_scaled_bias_by_channel.csv</code>     | <code>08_i7_scaled_validation.py</code>    |
| <code>i7_scaled_normal_precision.csv</code>    | <code>08_i7_scaled_validation.py</code>    |
| <code>i7_freeze_latency.csv</code>             | <code>09_i7_freeze_latency.py</code>       |
| <code>i7_regime_fp.csv</code>                  | <code>10_i7_regime_fp.py</code>            |
| <code>i7_deadband.csv</code>                   | <code>11_i7_ablations.py</code>            |
| <code>i7_freeze_quiescent.csv</code>           | <code>11_i7_ablations.py</code>            |
| <code>i7_intel_lab_operating_points.csv</code> | <code>12_i7_intel_lab_realfaults.py</code> |
| <code>i7_intel_lab_fault_types.csv</code>      | <code>12_i7_intel_lab_realfaults.py</code> |
| <code>i7_lbnl_bias.csv</code>                  | <code>13_i7_lbnl_bias.py</code>            |
| <code>i5_reunanen_skab.csv</code>              | <code>14_i5_reunanen_headtohead.py</code>  |
| <code>tsb_ad_fullrun.csv</code>                | <code>06_tsb_ad_fullrun.py</code>          |

526 *Artifact map.* Table 11 lists, for each committed CSV, the script that pro-  
527 duces it; the table and figure captions in Section 5 name the CSV each draws  
528 from, closing the loop from raw data to typeset result.

529 The full procedure—environment, data acquisition, per-artifact commands,  
530 and figure rendering—is documented in `publications/I7_fault_typing/REPRODUCE.md`  
531 at the same tag.

## 532 Additional information

533 The classifier and all evaluation artifacts are openly accessible on GitHub  
534 at <https://github.com/MarekWadinger/adaptive-interpretable-ad> (pinned  
535 at git tag `i7-fault-typing-v1`; see Section 8).

## 536 CRediT authorship contribution statement

537 **Marek Wadinger:** Conceptualization; Data curation; Formal analysis;  
538 Investigation; Methodology; Software; Validation; Visualization; Writing -  
539 original draft; and Writing - review & editing. **Michal Kvasnica:** Concep-  
540 tualization; Funding acquisition; Project administration; Resources; Super-  
541 vision; Validation.

## 542 Declaration of Competing Interest

543 The authors declare that they have no known competing financial inter-  
544 ests or personal relationships that could have appeared to influence the work  
545 reported in this paper.

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